15745 Project Milestone Report

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Project Title. Eager-Mode DNN Program Compilation with MagPy in TVM

Major Changes. The original goals in our project proposal are:

100% goal We can reimplement MagPy and match the evaluation results as in the MagPy paper.

75% goal We can reimplement MagPy, but the instantiation success rate is less than what MagPy paper reports. There are a few possible reasons that can lead to this result (for example, TVM lacks some operator support and thus cannot represent a ML model).

125% goal We can be better than what reported in the MagPy paper in either metric. If there is a chance we can leverage TVM for better optimizations, then we may be able to achieve better runtime performance.

The PyTorch model import workflow of the MagPy system contains two major steps. Firstly, it dynamically traces the model with the given input tensors, and generates a torch FX graph of the model. Secondly, it compiles the FX graph with the torch compiler.

Our initial plan is to reimplement MagPy to directly generate the graph-level IR Relax in TVM. After tracing the system code, we found that the MagPy implementation is highly coupled with the FX graph data structure. We estimate that there will be huge engineering effort if we are going to construct Relax IR directly from the model tracer in MagPy.

Therefore, we changed our plan to implement an IR translator, which translates constructed torch FX graphs to corresponding Relax IR, in which way we can also leverage TVM to optimize and compile the dynamic models imported by MagPy.

The goals above remain the same, except that the way of reimplementation changes from "constructing Relax IR directly in Magpy" to "translating the torch FX graphs generated by MagPy to Relax IR in TVM".

Accomplishment So Far. We have implemented a basic IR translator and can compile some basic example models imported by MagPy with TVM correctly.

Meeting the Milestone? Yes. The milestone of "implement MagPy with TVM" is met given we can compile and run example models correctly.

Revised Schedule. The schedule does not need much revision. We plan to evaluate the system with the benchmark provided by the MagPy paper's artifact evaluation. We might need to fix or enhance our translator if we find any model in the benchmark that is not yet fully supported.

Resources Needed. We have all the needed resources for the project.